of suitable wavelength (or frequency) is called **photoelectric effect**. Although these electrons are not different from all other electrons, it is customary to call them as **photoelectrons** and the corresponding current as **photoelectric current or photo current**.

Metals like cadmium, zinc, magnesium etc show photoelectric emission with ultraviolet light while some alkali metals lithium, sodium, caesium respond well even to larger wavelength radiation like visible light. The materials which eject photoelectrons upon irradiation of electromagnetic wave of suitable wavelength are called **photosensitive materials**.

8.2.2 Effect of intensity of incident light on photoelectric current

Experimental setup

The apparatus shown in Figure 8.8 is employed to study the phenomenon of photoelectric effect in detail. *S* is a source of electromagnetic waves of known and variable frequency *v* and intensity *I*. *C* is the cathode (negative electrode) made up of photosensitive material and is used to emit electrons. The anode (positive electrode) *A* collects the electrons emitted from *C*. These electrodes are kept in an evacuated glass envelope with a quartz window that permits the passage of ultraviolet and visible light.

The necessary potential difference between *C* and *A* is provided by high tension battery *B* which is connected across a potential divider arrangement *PQ* through a key *K*. *C* is connected to the centre terminal while *A* to the sliding contact *J* of the potential divider. The plate *A* can be maintained at a desired positive or negative potential with respect to *C*. To measure both positive and negative potential of *A* with respect to *C*, the voltmeter is designed to have its zero marking at the centre and is connected between *A* and *C*. The current is measured by a micro ammeter μA connected in series.

If there is no light falling on the cathode *C*, no photoelectrons are emitted and the microammeter reads zero. When ultraviolet or visible light is allowed to fall on *C*, the photoelectrons are liberated and are attracted towards anode. As a result, the

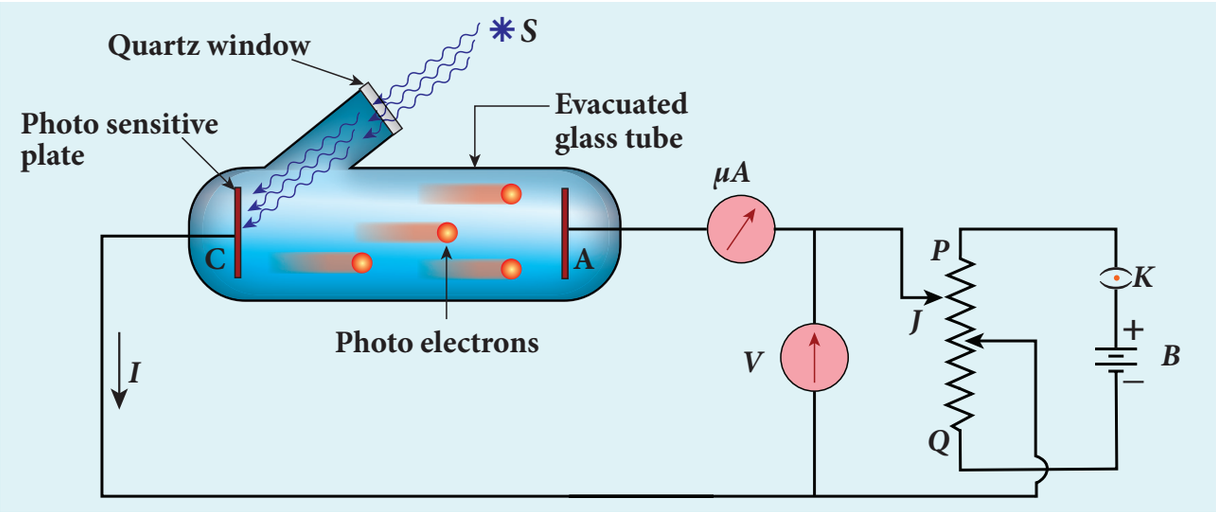


Figure 8.8 Experimental setup for the study of photoelectric effect

XII_Physics_Vol-2_UNIT-8_EM.indd 113 03-03-2022 15:20:21